

INTRODUCTION TO PROOF WRITING (MATH 300B)

University of Washington, Summer 2025

Instructor:	Mallory Dolorfino (they)	Time:	MWF 10:50am – 11:50am
Email:	mallod2@uw.edu	Place:	SIG 228

TA/Grader: Varun Shah (he)

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Office Hours (subject to change based on student feedback):

- Wednesdays, 12:00pm-1:00pm in the Math Study Center with Mallory
- Thursdays, 11am-12pm in the Math Study Center with Varun
- Fridays, 9:30am-10:30am in PDL C-404 with Mallory

Main References: This is a non-comprehensive list of several textbooks that may be useful throughout the course, though you will not be required to do any reading.

- Clive Newstead, *An Infinite Descent into Pure Mathematics*. Freely available [here](#)
- Matthew Conroy and Jennifer Taggart, *An Introduction to Mathematical Reasoning*. Freely available [here](#)
- Ted Sundstrom, *Mathematical Reasoning: Writing and Proof*. Freely available [here](#)

Course Objectives: By the end of this course, you will have the skills to correctly piece together mathematical definitions and assumptions to rigorously prove mathematical statements. Moreover, you will be able to communicate your proofs to others in an effective, intelligible way.

Course Content: Math 300 emphasizes mathematical arguments and the writing of proofs. Topics include, but are not limited to, the following:

- Logic and mathematical language
- Direct, indirect, and contrapositive proofs
- Existence and uniqueness proofs
- Elementary set theory
- Functions
- Cardinality
- Elementary number theory
- Elementary abstract algebra
- Typesetting using $\text{\LaTeX}$

Grading Policy: All assignments, including midterms and the final, will be graded using an **E** (excellent), **S** (satisfactory), **N** (needs work) grading system. The following rubric will be used to assess each assignment:

Excellent (E)	1 or 2 insignificant mistakes
Satisfactory (S)	> 2 insignificant mistakes, or 1 significant mistake
Needs Work (N)	> 3 insignificant mistakes, or > 1 significant mistakes

There will be 8 homework assignments and 2 midterms, for a total of 10 assignments before the final. Your grade before the final will be determined as follows:

4.0	≥ 9 Excellents
3.8	7-8 Excellents
3.6	6 Excellents, or 10 Satisfactories
3.4	5 Excellents, or 8 Satisfactories
3.2	≤ 4 Excellents

In addition, 10 Excellents warrants a 4.0 **regardless of your grade on the final**, though you still must take the final and demonstrate some considerable effort. Similarly, 3 or more Needs Works warrants a failing grade, **regardless of your grade on the final**. Otherwise, your grade on the final results in the following changes to your grade before the final:

Excellent (E)	+ .2
Satisfactory (S)	+ 0
Needs Work (N)	- .2

Redos: You may redo 5 assignments (including the midterms) before the final. An adequate redo attempt increases your original grade on the assignment to the next level only, and you cannot redo the same assignment twice. For example, if you redo an assignment that was initially graded as a **N**, a redo improves this grade to an **S**, but no higher. An adequate redo attempt consists of (1) attempting to correct each mistake and (2) giving a short explanation of why your original work was incorrect. Note that though it is ideal that your redo attempt is completely correct, it will be adequate regardless – the most important part of the redo is the **reflection on your original mistakes and a genuine attempt to fix them**.

Passing Threshold: A final grade of 2.0 or higher in this course indicates that you are prepared to independently prove a statement for one of the subsequent courses (e.g. Math 327) using any of the proof techniques covered in this course (contradiction, induction, etc).

Important Dates:

Midterm #1 July 18, 2025, during class time
 Midterm #2 August 8, 2025, during class time
 Final Exam August 22, 2025, during class time

Weekly Schedule:

- Mondays and Wednesdays will be a traditional lecture given by me
- Fridays will be focused on group problem solving trickier problems
- Homework will be due every week starting week 2 (including midterm and final weeks) at 11:59pm – see the next section for more details on homework

Homework:

- Homework assignments will be posted on Wednesday of the week before they are due. Each will consist of 4 exercises – you are expected to attempt each problem. Failure to do so will result in an automatic **N**
- You will turn in homework on Gradescope, which will be accessible through the course Canvas page
- Homework will be graded based on mathematical and logical correctness, using the general rubric given above
- Though your homework grades will be based only on mathematical correctness, Varun and I will make many comments about **how you write your proofs**. Mathematical proofs are written to communicate with others, so it is imperative that your writing be clear and understandable
- Failure to turn in a homework assignment before the end of the course results in an **automatic failure**. Late homework will be accepted if you contact me at least 48 hours in advance, and the highest original grade on a late assignment is an **S**

Additional Requirements: You will be required to use $\text{\LaTeX}$ to typeset at least 3 of your homework assignments. Failure to do so results in the lowering of one of your highest grades to the next lowest level. You may use this [online workspace and editor](#) to create and organize your $\text{\LaTeX}$ submissions, and you may duplicate this [template](#) to get started. In addition, this [YouTube video](#) and this [tutorial](#) will help you learn the basics of $\text{\LaTeX}$!

Attendance Policy: Regular attendance is essential and expected. You are allowed **3 unexcused absences**, no questions asked. Any additional absence will result in the lowering of one of your highest grades to the next lowest grade. For example, if your highest grade is an **S**, an additional absence lowers that grade to a **N**. **Please talk to me if you have any extenuating circumstances, and we can try to work out a solution.**

Illness: If you are not feeling well enough to attend any exam, for the safety of your colleagues and yourself, please stay home and rest. To ensure we can give you the appropriate accommodations, please inform me via email within 48 hours of the exam.

If you are sick and need to miss a class that would be your 3rd absence or more, please send me an email within 48 hours of the class, and we can find a suitable accommodation.

Communication: All communication for the class should be done through me, preferably via in person discussions. I will stay for 10 minutes after every class, and I am happy to schedule individual meetings if needed. If you must email, I will reply within 24 hours, except for on the weekends. If you do email me, it should be primarily to tell me if you must miss a class, or to schedule a personal meeting.

Academic Honesty: You are highly encouraged to work on homework assignments with your classmates, with me, and with Varun. It is recommended but not required that you cite on each assignment who you worked with, but failure to do so will not result in any negative consequences.

Artificial Intelligence: It is highly recommended that you **completely avoid** using any kind of artificial intelligence during this course. Doing so will greatly harm your ability to write proofs now and in the future. Writing proofs is creative and individual – like any other skill, it requires a lot of practice and hard work to get good at – using artificial intelligence may make things easier in the moment, but it will prohibit you from learning how to be creative and thorough while problem solving. As there is no fair way to determine whether student work is AI-written or not, there will be no consequences for using AI on your assignments, but Varun and I trust that you will not use it for the duration of the course.

Class Norms: I am a firm believer in Federico Ardila's axioms:

1. Mathematical potential is equally present in different groups, irrespective of geographic, demographic, and economic boundaries
2. Everyone can have joyful, meaningful, and empowering mathematical experiences
3. Mathematics is a powerful, malleable tool that can be shaped and used differently by various communities to serve their needs
4. Every student deserves to be treated with dignity and respect

I use these axioms to inform my teaching practices and beliefs, and I want all of you to feel safe, respected, and capable during this course. Towards this end, we will create a running list of [class norms](#), which you can edit at any time. These norms will be used to ensure that every student feels empowered and respected for the duration of this course. You may call on these class norms to speak out if you see any of your classmates being disrespectful, or you may use it to report any problems to me. I have already started the list of norms, but we will edit and add to them on the first day of the class, and we can reassess them often as the course goes on.

Practical Advice: To stay on top of your studies in this class, I recommend doing the following:

- Start your homework as soon as it is available (even if briefly looking over the homework)
- Start studying for an exam about 10 days prior
- Treat every part of our class as a learning opportunity, including exams, and especially redos! Ask yourself what went well and how you might do things differently next time
- Come to class curious and unafraid to ask questions! If you have a question or curiosity, it is likely that others feel the same, and asking questions helps you learn
- Come to office hours with Varun and me as much as you can, even if you do not have any specific questions. We are more than happy to help with homework or exam prep, and we are even happy to discuss math content not relating to the course

Disability Resources: Your experience in this class is important to me. It is the policy and practice of the University of Washington to create inclusive and accessible learning environments consistent with federal and state law. If you have already established accommodations with Disability Resources for Students (DRS), please activate your accommodations via [myDRS](#) so we can discuss how they will be implemented in this course.

If you have not yet established services through DRS, but have a temporary health condition or permanent disability that requires accommodations (conditions include but are not limited to: mental health, attention related, learning, vision, hearing, physical, or health impacts), contact DRS directly to set up an Access Plan. DRS facilitates the interactive process that establishes reasonable accommodations. Contact DRS [here](#).

Religious Accommodations: Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UW's policy, including more information about how to request an accommodation, is available [here](#). Accommodations must be requested within the first two weeks of this course using the [Religious Accommodations Request form](#).

SafeCampus: Call SafeCampus at 206-685-7233 anytime – no matter where you work or study – to anonymously discuss safety and well-being concerns for yourself or others. SafeCampus's team of caring professionals will provide individualized support, while discussing short-term and long-term solutions and connecting you with additional resources when requested.

Title IX: UW, through numerous policies, prohibits sex-based and gender-based violence and harassment, and we expect students, faculty, and staff to act professionally and respectfully in all work, learning, and research environments. For support, resources, and reporting options related to sex-based or gender-based violence or harassment, I will connect you (or the person who experienced the conduct) with resources and individuals who can best provide support and options. You can also access those resources directly:

- Confidential: Confidential advocates will not share information with others unless given express permission by the person who has experienced the harm or when required by law
- Private and/or anonymous: SafeCampus provides consultation and support and can connect you with additional resources if you want them. You can contact SafeCampus anonymously or share limited information when you call

Please note that some senior leaders and other specified employees have been identified as "Officials Required to Report." If an Official Required to Report learns of possible sex-based or gender-based violence or harassment, they are required to call SafeCampus and report all the details they have in order to ensure that the person who experienced harm is offered support and reporting options.